

Discrete Systems: State and Transitions

- A state machine is a tuple (S, I, T, s_0) , where S is a finite state set.
- Deterministic transition: each state-input pair has at most one successor.
- Invariant proof: establish $P(s_0)$, then show $P(s)$ implies $P(T(s, i))$.
- Reachability asks whether a target state occurs on any valid input sequence.
- Example: a two-bit counter advances $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$.

Probability Refresher

- Conditional probability: $P(A | B) = P(A \text{ and } B) / P(B)$, for $P(B) > 0$.
- Bayes rule: $P(A | B) = P(B | A) P(A) / P(B)$.
- Linearity: $E[X + Y] = E[X] + E[Y]$, without requiring independence.
- Variance: $\text{Var}(X) = E[X^2] - E[X]^2$.
- For independent X and Y , $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

Asymptotic Analysis

- Big-O is an eventual upper bound; Big-Omega is an eventual lower bound.
- A geometric series $1 + r + \dots + r^k$ equals $(r^{k+1} - 1) / (r - 1)$.
- Binary search satisfies $T(n) = T(n/2) + O(1)$, so $T(n) = O(\log n)$.
- Merge sort satisfies $T(n) = 2T(n/2) + O(n)$, so $T(n) = O(n \log n)$.
- Always state the input measure and distinguish worst, average, and amortized cost.